

CV Date

04/06/2026

Part A. PERSONAL INFORMATION

First Name	David		
Family Name	Rodríguez Martínez		
Sex	Male	Date of Birth	03/10/1990
ID number Social Security, Passport			
URL Web	https://drodriguezsr.github.io/		
Email Address	david.rm@uma.es		
Open Researcher and Contributor ID (ORCID)	0000-0003-4817-9225		

A.1. Current position

Job Title	Beatriz Galindo Distinguished Professor		
Starting date	2024		
Institution	Universidad de Málaga		
Department / Centre	Departamento de Ingeniería de Sistemas y Automática / Escuela de Ingenierías Industriales		
Country	Spain	Phone Number	(0034) 951952332
Keywords	Space Robotics; Perception; Autonomous Navigation; Mobility; Terramechanics; Localization & Mapping; Advanced Sensing;		

A.2. Previous positions (Research Career breaks included)

Period	Job Title / Name of Employer / Country
2025 - 2025	[career break] Paternity leave form 17/07/2025-03/12/2025
2023 - 2024	Research Scientist (postdoc) / École Polytechnique Fédérale de Lausanne (EPFL) / Switzerland
2020 - 2023	Systems Engineer / École Polytechnique Fédérale de Lausanne (EPFL) / Switzerland
2018 - 2020	NPI Researcher / European Space Agency / Holland
2017 - 2020	PhD Candidate / Tohoku University / Japan
2018 - 2018	Visiting Scientist / German Aerospace Agency / Germany
2016 - 2016	Mechanical Engineer Trainee / European Space Agency / Holland
2014 - 2015	Research Scholar / West Virginia University / United States of America

A.3. Education

Degree/Master/PhD	University / Country	Year
PhD in Engineering	Tohoku University / Japan	2020
MSc. in Space Studies	International Space University / France	2016
Ingeniero Industrial Especialidad Mecánica	Universidad Carlos III de Madrid / Spain	2015

Part B. CV SUMMARY

Since December 2024, I have been appointed as a Beatriz Galindo Distinguished Professor at the University of Malaga, where I co-lead the Space Robotics Lab. My work intersects robotics, advanced mechanical design, quantum optics, computer vision, and space. My research has pioneered advancements in faster lunar locomotion and single-photon vision, with the goal of ultimately enabling planetary robots to autonomously explore some of the most extreme environments in our solar system. Over more than ten years, I founded initiatives and led research projects in academic institutions and public organizations based in Europe, USA, and Japan, raising over €1.1M in competitive research grants (as PI/co-PI) from international, national, and regional public and private institutions. I have been honored with prestigious personal grants and awards, including the ESA-ISU sponsorship, the Japanese MEXT fellowship, Tohoku University's GP Mech award, ESA's Networking Partnering Initiative, and, more recently, the Beatriz Galindo fellowship from the Spanish Ministry of Science, Innovation, & Universities. I have been involved in the organization of the International Conference on Space Robotics (iSpaRo) as co-chair in 2024 and member of the program committee in 2025 and 2026. I have also organized multiple workshops at ICRA 2023, 2024, and 2026, and ERF 2024, as well as several smaller workshops at other venues. Throughout my career, I have been invited to present my work in embassies, universities, and more than a dozen events, conferences, and symposiums. I have been Associate Editor in the last three editions of the iSpaRo conference, guest Editor for JINT in 2023, and I actively review papers for a number of journals and conferences including RAL, RAM, Journal of Terramechanics, ICRA, AeroConf, and ISTVS, to name a few. Aside from academia, I served as a Systems Engineer, later Principal Scientist, at EPFL's Space Center from 2020-2022, where I founded and directed the Lunar Hub aimed at advancing lunar R&D in Switzerland. During this period, I launched the Lunar Reconnaissance Drone project and managed over €3M in funding, contributing to innovation/ tech transfer projects including ESA-ADRIOS, CAPSYS, RELNAV, led by ClearSpace, and NESTS, led by Ariane Group. I worked as an Associate Researcher for ESA from 2018-2020 and as a Visiting Scientist for DLR in 2019. I have supervised (directed or co-directed) 40+ undergrad/grad student projects and coordinated initiatives involving more than 200 students. I am currently co-supervising 1 PhD thesis at the University of Malaga and have indirectly supported the work of 5 PhDs at Tohoku University and EPFL. I currently teach undergraduate courses on the Fundamentals of Control (GIERM209) and Product Automation (GIDIDP311) at UMA's School of Industrial Engineering, and I am recurrently invited to give a Lecture Series on Space Robotics at the International Space University since 2025. Previously, I taught and directed graduate courses at EPFL on Concurrent Engineering of Space Missions (ENG411) and Spacecraft Design and Systems Engineering (EE584), with around 60 students enrolled every year. Currently, I chair the HERMES committee, serve on euRobotics TGs on Space Robotics, Autonomous Navigation, and Perception, and on IEEE RAS TCs on Space Robotics, Computer & Robot Vision, and Robot Learning; and I participated as a member in NASA SSERVI's Swiss Affiliate Network.

Part C. RELEVANT ACCOMPLISHMENTS

C.1. Most important publications in national or international peer-reviewed journals, books and conferences

- 1 **Scientific paper.** **D. Rodríguez-Martínez** (CA); D. van der Meer; J. Song; A. Bera; C. J. Pérez-del-Pulgar; M. A. Olivares-Mendez. 2026. SPICE-HL3: Single-Photon, Inertial, and Stereo Camera dataset for Exploration of High-Latitude Lunar Landscapes. Scientific Data 12, 374. Springer Nature. DOI: 10.1038/s41597-026-06668-8.
- 2 **Scientific paper.** R. Tonasso; D. Tataru; H. Rauch; V. Pozsgay; T. Pfeiffer; E. Uythoven; **D. Rodríguez-Martínez** (CA). 2024. A lunar reconnaissance drone for cooperative exploration and high-resolution mapping of extreme locations. Acta Astronautica. Elsevier. 218, pp.1-17. DOI: 10.1016/j.actaastro.2024.02.006
- 3 **Scientific paper.** **D. Rodríguez-Martínez** (CA); K. Uno; K. Sawa; G. Kudo; G. Hernan Diaz; A. Umemura; S. Santra; K. Yoshida. 2024. Enabling Faster Locomotion of Planetary Rovers with a Mechanically-Hybrid Suspension. IEEE Robotics and Automation Letters (RA-L). IEEE. 9-1, pp.619-626. DOI: 10.1109/LRA.2023.3335769
- 4 **Scientific paper.** **D. Rodríguez-Martínez** (CA); K. Yoshida. 2017. High-speed lunar rovers. ROOM: The Space Journal. 2-12, pp.54-56.
- 5 **Scientific paper.** **D. Rodríguez-Martínez** (CA); M. Van Winnendael; K. Yoshida. 2019. High-speed mobility in planetary surfaces: a technical review. Journal of Field Robotics. Wiley. 36-8, pp.1436-1455. DOI: 10.1002/rob.21912

C.2. Conferences and meetings

- 1 A. Martínez-Petersen; L. Gerdes; **D. Rodríguez-Martínez**; C. J. Pérez-del-Pulgar. High fidelity 3D reconstruction for planetary exploration. IEEE Conference on Artificial Intelligence (CAI). Granada, Spain. 2026.
- 2 **D. Rodríguez-Martínez**; C. Pérez del Pulgar. Fast vision in the dark: A case for single-photon imaging in planetary navigation. 18th Symposium on Advanced Space Technologies in Robotics and Automation (ASTRA). Leiden, The Netherlands. 2025.
- 3 C. Pérez del Pulgar; **D. Rodríguez-Martínez**; L. Gerdes; A. J. García Cerezo. The role of robotics in planetary exploration, contributions from the University of Málaga. VIII Congreso de Ciencias Planetarias (CPESS-8). Málaga, Spain. 2025.
- 4 A. Guerrieri; **D. Rodríguez-Martínez**; E. Charbon. A Dynamically Reconfigurable Single Board Computer for High-Dynamic Range Space Cameras. IEEE Space Mission Challenges for Information Technology / Space Computing Conference (SMC-IT/SCC). Mountain View, CA. 2024.
- 5 T. Manteaux; **D. Rodríguez-Martínez**; R. Thilak Rajan. RAPF: Efficient path planning for lunar microrovers. International Conference on Space Robotics (iSpaRo). Luxembourg. 2024.
- 6 K. Sawa; K. Uno; G. Kudo; K. Yoshida; **D. Rodríguez-Martínez**. Development and experimental evaluation of a suspension mechanism for a high-speed lunar rover. Robotics and Mechatronics Conference (ROBOMECH). Japan Society of Mechanical Engineers (JSME). Nagoya, Japan. 2023.
- 7 **D. Rodríguez-Martínez**; T. Pfeiffer; E. Uythoven; V. Pozsgay; R. Tonasso; E. David; J-P. Kneib. Design of a lunar reconnaissance drone for exploration and mapping of extreme, hardly accessible locations. Space Resources Week. European Space Resources Innovation Centre. Luxembourg. 2023.
- 8 V. Pozsgay; **D. Rodríguez-Martínez**; J-P. Kneib. A lunar reconnaissance drone mission concept for mapping and characterizing polar regions. Lunar Polar Volatiles Conference (LPVC). Lunar and Planetary Institute (LPI). Boulder, CO. 2022.
- 9 T. Pfeiffer; E. Uythoven; **D. Rodríguez-Martínez**; H. Koizumi; J-P. Kneib. Feasibility study and preliminary design of a lunar reconnaissance drone. Lunar Surface Innovation Consortium (LSIC) Spring Meeting. The Johns Hopkins University Applied Physics Laboratory. Laurel, MD. 2022.
- 10 K. Nakagoshi; **D. Rodríguez-Martínez**; K. Yoshida. A new single-wheel test bed for

fast-moving planetary robots. Aerospace Europe Conference. French National Aeronautical and Astronautical Association (3AF). Bordeaux, France. 2020.

- 11 **D. Rodríguez-Martínez**; F. Buse; M. Van Winnendael; K. Yoshida. The effects of increasing velocity on the tractive performance of planetary rovers. 15th ISTVS Conference. International Society for Terrain-Vehicle Systems (ISTVS). Prague, Czech Republic. 2019.
- 12 P. Iliffe; S. Kaethler; E. Xu; **D. Rodríguez-Martínez**, et al. Planetary protection and the search for life on the icy moons of the Solar System: A technology roadmap. 67th International Astronautical Congress (IAC). International Astronautical Federation (IAF). Guadalajara, Mexico. 2016.

C.3. Research projects and contracts

- 1 **Project**. Single-Photon Imaging to Explore the Moon. European Space Agency (ESA). PI: David Rodríguez Martínez (Universidad de Málaga). 01/03/2026-31/12/2026. 20.000 €.
- 2 **Project**. (INSIGHT) Hybrid Intelligence for Space Robotics: Merging Classical and Learned Methods for Robust and Interpretable GNC. PI: Carlos Pérez del Pulgar (Universidad de Málaga). 01/09/2025-31/08/2028. 105.000 €. My role: Research Team Member.
- 3 **Project**. (ECHO2) End-to-End Control for Handling Operations in Orbit. European Space Agency (ESA). PI: David Rodríguez Martínez (Universidad de Málaga). 01/10/2025-30/09/2027. 140.000 €.
- 4 **Project**. Monocular SPAD camera for enhanced vision in complex and uncertain environments. armassuise Science & Technology (S+T) Program, Federal Department of Defense, Civil Protection and Sport (DDPS). PI: David Rodríguez Martínez. (École Polytechnique Fédérale de Lausanne (EPFL)). 01/04/2024-31/12/2024. 100.000 €.
- 5 **Project**. Quantum LiDAR for NLOS applications. armassuise Science & Technology (S+T) Program, Federal Department of Defense, Civil Protection and Sport (DDPS). PI: David Rodríguez Martínez. (École Polytechnique Fédérale de Lausanne (EPFL)). 01/03/2024-15/11/2024. 86.000 €.
- 6 **Project**. Pre-study on SPAD-based LiDAR Direct Time-of-flight (DToF) for occluded perception. Armasuisse Science & Technology (S+T) Program, Federal Department of Defense, Civil Protection and Sport (DDPS). PI: Edoardo Charbon. (École Polytechnique Fédérale de Lausanne (EPFL)). 01/04/2023-31/12/2023. 37.000 €. My role: Scientific Coordinator
- 7 **Project**. (DRAGONFLY) A 1 Mpx SPAD camera with real-time on-chip computation for space applications. armasuisse Science and Technology (S+T), Federal Department of Defense, Civil Protection and Sport (DDPS). PI: David Rodríguez Martínez. (École Polytechnique Fédérale de Lausanne (EPFL)). 10/11/2022-31/12/2023. 140.000 €.
- 8 **Project**. (NESTS) New European Space Transportation Solutions. European Space Agency (ESA) FLPP. PI: Emmanuelle David. (École Polytechnique Fédérale de Lausanne (EPFL)). 01/03/2021-31/07/2021. My role: Project Manager for EPFL.

C.4. Participation in technology/knowledge transfer activities

- 1 **Contract**. ADRIOS ClearSpace Service Phase 1B/2 European Space Agency (ESA). PI: Jean-Paul Kneib. 01/05/2021-01/05/2023. 1.175.000 €. My role: Project Manager for EPFL.
- 2 **Contract**. Capture system concept validation. Swiss Innovation Agency (Innosuisse). PI: Jean-Paul Kneib. 01/06/2020-31/10/2021. 607.900 €. My role: Project Manager for EPFL.
- 3 **Contract**. Impulse-SpaceRelNav: Relative navigation technologies for failed satellite removal Swiss Innovation Agency (Innosuisse). P I : Jean-Paul Kneib. 01/11/2019-01/11/2020. 1.536.759 €. My role: Project Manager for EPFL.
- 4 **Contract**. High-speed lunar locomotion. Tohoku University Graduate School of Engineering (GPMech) Grant. P I : David Rodríguez Martínez. 01/01/2018-01/10/2020. 12.500 €
- 5 **Contract**. Study, analysis and development of a high-speed locomotive system for an improved-mobility lunar prospecting rover European Space Agency (ESA) Networking-Partnering Initiative (NPI) Research Programme. PI: David Rodríguez Martínez. 01/01/2018-01/07/2020.